1. What are clustered and non-clustered indexes?

Answer:

- **Clustered Index**: Data is physically stored in the order of the index. Only one per table. The leaf nodes contain the actual data pages.
- **Non-Clustered Index**: Separate structure that holds index key values and pointers to the data rows. Multiple allowed per table.
- **Example**: Primary Key by default creates a clustered index.

---

2. What are Common Table Expressions (CTEs)?

Answer:

- A CTE is a temporary result set defined using the WITH keyword.
- Useful for **recursive queries** and improving readability of complex queries.
- Example:

```
WITH EmployeeCTE AS (
  SELECT EmployeeID, ManagerID, Name
  FROM Employees
  WHERE ManagerID IS NULL
  UNION ALL
  SELECT e.EmployeeID, e.ManagerID, e.Name
  FROM Employees e
  INNER JOIN EmployeeCTE c ON e.ManagerID = c.EmployeeID
)
SELECT * FROM EmployeeCTE;
```

---

3. Explain ACID properties in SQL Server.

Answer:

- **Atomicity**: All operations in a transaction succeed or none do.
- **Consistency**: Data must remain in a valid state after the transaction.
- **Isolation**: Concurrent transactions are isolated from each other.
- **Durability**: Once committed, data survives failures.

4. What are SQL Server isolation levels?

Answer:

- **Read Uncommitted**: Allows dirty reads.
- **Read Committed (default)**: Prevents dirty reads, but allows non-repeatable reads.
- **Repeatable Read**: Prevents dirty and non-repeatable reads.
- **Serializable**: Strictest, prevents phantom reads too.
- **Snapshot**: Uses row versioning to avoid blocking.

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5. What's the difference between DELETE, TRUNCATE, and DROP?

Answer:

- **DELETE**: Removes rows, logs each row deletion, can use WHERE. Slower.
- **TRUNCATE**: Removes all rows, deallocates pages, minimal logging. Faster. No WHERE.
- **DROP**: Deletes the table structure entirely.

---

6. What are SQL Server execution plans?

Answer:

- Execution plans show how SQL Server executes a query.
- **Types**:
  - **Estimated Plan** (before execution).
  - **Actual Plan** (after execution).
- Useful for identifying **missing indexes, table scans, costly operators**.
- Accessed via SET SHOWPLAN_ALL ON or in SSMS (Ctrl + M).

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7. How do you improve query performance in SQL Server?

Answer:

- Create proper **indexes** (covering indexes, filtered indexes).
- Avoid SELECT *, retrieve only needed columns.
- Use **joins** efficiently instead of subqueries.
- Keep statistics updated (UPDATE STATISTICS).
- Avoid **cursors**, use set-based operations.

- Use **execution plans** to tune queries.

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8. What are stored procedures and why use them?

Answer:

- Stored procedures are precompiled SQL statements stored in the database.
- **Advantages:**
    - Improved performance (compiled once, cached).
    - Encapsulation of business logic.
    - Better security (permissions at procedure level).
    - Reusability.

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9. What are SQL Server triggers and their types?

Answer:

- Triggers are special stored procedures that execute automatically on INSERT, UPDATE, DELETE.
- **Types:**
    - **AFTER Trigger**: Fires after the action completes.
    - **INSTEAD OF Trigger**: Replaces the action with custom logic.
- Example use case: Auditing changes to a table.

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10. What's the difference between a primary key, unique key, and foreign key?

Answer:

- **Primary Key**: Ensures uniqueness, no NULLs, only one per table.
- **Unique Key**: Ensures uniqueness, allows one NULL, multiple per table.
- **Foreign Key**: Ensures referential integrity, links to primary/unique key in another table.